

Deciding existence of equilibria in Fisher markets with earning and utility limits

ChatGPT (gpt-5.6-sol)*

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Abstract

We prove that deciding whether a Fisher market with utility-capped buyers and earning-capped goods admits a thrifty-and-modest equilibrium is strongly NP-complete. Membership in NP is not immediate, because an equilibrium could, a priori, require irrational prices. We show that after guessing the zero prices, the binding sides of the utility and earning caps, the maximum-bang-per-buck equalities, and the allocation supports, the equilibrium conditions form a rational linear system. A positive-slack linear program then yields a polynomial-bit rational equilibrium whenever an equilibrium exists.

For hardness, we give a reduction from EXACT COVER BY 3-SETS. The construction uses an anchor gadget that fixes a reference price at 24, an element gadget that forces two element prices to 24 and demands exactly 12 dollars of token money for every element, and a set gadget with exactly two possible price states. A set in its high state sends one 12-dollar token to each of its three elements; a set in its low state sends none. Market clearing therefore holds precisely when the high sets form an exact cover. All numerical parameters in the constructed instance belong to fixed finite sets, establishing strong NP-hardness.

1 The open problem

This paper addresses an open problem posed by Jugal Garg, Martin Hoefer, Kurt Mehlhorn in *Satiation in Fisher Markets and Approximation of Nash Social Welfare* [1] (Mathematics of Operations Research, 2023), Section 5 (Future Directions), page 32.. The website catalogue entry is problem 57 (W2965273013_p0) of the [Open Problems in Operations Research](#) collection.

Given a Fisher market instance specified by B, G and nonnegative parameters (u_{ij}) , (m_i) , (c_i) , and (d_j) , determine whether there exists a thrifty and modest equilibrium (x, p) satisfying:

- (i) $p \geq 0$; (ii) each x_i is an MBB and modest demand bundle at p ; (iii) $x_j \leq \min\left(1, \frac{d_j}{p_j}\right)$; (iv)

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Equivalently, decide the existence (and, if desired, compute one) of such an equilibrium for markets with both earning caps (d_j) and utility caps (c_i).

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Market model

There is a finite set I of buyers and a finite set J of divisible goods. Every good has one unit of physical supply. Good j has an earning cap $d_j > 0$. Buyer i has a budget $m_i \geq 0$, a utility cap $c_i > 0$, and nonnegative linear utility coefficients u_{ij} .

For a bundle $x_i \in \mathbb{R}_+^J$, define its raw and capped utilities by

$$v_i(x_i) = \sum_{j \in J} u_{ij} x_{ij}, \quad U_i(x_i) = \min\{c_i, v_i(x_i)\}.$$

At prices $p \in \mathbb{R}_+^J$, the budget-feasible set of buyer i is

$$\mathcal{B}_i(p) = \left\{ y \in \mathbb{R}_+^J : \sum_{j \in J} p_j y_j \leq m_i \right\}.$$

A bundle is *thrifty* if it first maximizes capped utility and, among all utility-maximizing bundles, minimizes expenditure. The thrifty demand correspondence is therefore

$$\mathcal{D}_i(p) = \arg \min_y \left\{ p \cdot y : y \in \arg \max_{z \in \mathcal{B}_i(p)} U_i(z) \right\}.$$

A bundle x_i is *modest* if

$$v_i(x_i) \leq c_i.$$

The unilateral demand problem above is unrestricted: a buyer's alternative bundles are not individually constrained by the market's unit supplies. Unit-supply constraints apply to the aggregate equilibrium allocation.

Definition 1 (Thrifty-and-modest equilibrium). A price vector $p \in \mathbb{R}_+^J$ and an allocation $x \in \mathbb{R}_+^{I \times J}$ form a *thrifty-and-modest equilibrium* if:

1. $x_i \in \mathcal{D}_i(p)$ and $v_i(x_i) \leq c_i$ for every buyer i ;
- 2.

$$\sum_{i \in I} x_{ij} \leq 1 \quad \text{for every } j \in J;$$

3. every good receives exactly its permitted revenue:

$$p_j \sum_{i \in I} x_{ij} = \min\{p_j, d_j\} \quad \text{for every } j \in J.$$

When $p_j > 0$, the last two conditions imply

$$\sum_i x_{ij} = \min \left\{ 1, \frac{d_j}{p_j} \right\}.$$

When $p_j = 0$, the revenue equation is vacuous, but the physical capacity constraint

$$\sum_i x_{ij} \leq 1$$

continues to apply.

The decision problem THRIFTY-AND-MODEST EQUILIBRIUM EXISTENCE asks whether a market given by rational data (u, m, c, d) admits such an equilibrium.

3 A demand characterization

For a buyer who values no free good, define the minimum price per unit of raw utility by

$$\beta_i(p) = \min_{j: u_{ij} > 0} \frac{p_j}{u_{ij}}.$$

Equivalently, $1/\beta_i(p)$ is the buyer's maximum bang per buck (MBB).

Lemma 1. *Fix prices p .*

1. *Suppose buyer i values no zero-price good. Then $\beta_i(p) > 0$, and every thrifty-and-modest bundle uses only goods satisfying*

$$p_j = u_{ij}\beta_i(p)$$

and has expenditure

$$b_i(p) = \min\{m_i, c_i\beta_i(p)\}.$$

2. *Suppose buyer i values at least one zero-price good. Then a thrifty-and-modest bundle has expenditure zero and receives raw utility exactly c_i , entirely from zero-price goods.*

Proof. In the first case, for every bundle $y \geq 0$,

$$p \cdot y = \sum_j p_j y_j \geq \beta_i(p) \sum_j u_{ij} y_j = \beta_i(p) v_i(y).$$

Equality is attainable using a good minimizing p_j/u_{ij} . Consequently, the greatest attainable raw utility under budget m_i is $m_i/\beta_i(p)$, and the greatest capped utility is

$$\min \left\{ c_i, \frac{m_i}{\beta_i(p)} \right\}.$$

The least expenditure attaining that utility is

$$\min\{m_i, c_i\beta_i(p)\}.$$

Equality in the displayed expenditure bound requires the bundle to use only maximum-bang-per-buck goods.

In the second case, buyer i can reach its finite utility cap at zero cost by taking a sufficient amount of a valued free good in its unilateral demand problem. Its maximum capped utility is therefore c_i , and the minimum expenditure attaining it is zero. Modesty then forces its raw utility to equal, rather than exceed, c_i . $\square$

4 Membership in NP

The subtle point in proving membership in NP is that merely guessing a real equilibrium is not an NP certificate. We must show that whenever an equilibrium exists, one exists with polynomial-bit rational prices and allocations.

Proposition 1. *The problem THRIFTY-AND-MODEST EQUILIBRIUM EXISTENCE belongs to NP.*

Proof. A buyer with $u_{ij} = 0$ for every j can be assigned the zero bundle and may be discarded. Assume, therefore, that every remaining buyer values at least one good.

We describe a polynomial-size discrete guess and a rational linear program for the corresponding equilibrium regime.

First, guess the set of zero-price goods and its complement,

$$Z = \{j : p_j = 0\}, \quad P = J \setminus Z.$$

For every buyer i , the guessed set Z determines whether i values a free good.

For each buyer who values no good in Z , guess:

- whether its active expenditure is budget-limited or cap-limited;
- one MBB witness $w_i \in P$ with $u_{iw_i} > 0$;
- the exact support $S_i \subseteq P$ of its positive spending.

The MBB witness is guessed independently of the spending support. This is essential for buyers with zero spending: their MBB variable must still equal the price per utility of an actual good.

For each buyer who values a good in Z , guess a valued free witness and the support of its free allocation.

For every positive-price good, guess whether the good sells its entire unit or reaches its earning cap.

Introduce variables

$$p_j \geq 0, \quad \beta_i \geq 0, \quad b_i \geq 0,$$

together with dollar-spending variables $y_{ij} \geq 0$ on positive-price goods and free-allocation variables $z_{ij} \geq 0$ on zero-price goods; here y_{ij} will represent the expenditure $p_j x_{ij}$.

For $j \in Z$, impose $p_j = 0$. For $j \in P$, impose

$$p_j \geq \delta, \quad 0 \leq \delta \leq 1.$$

Consider first a buyer i who values no good in Z . Impose

$$u_{ij}\beta_i \leq p_j \quad (j \in J)$$

and the witnessed equality

$$p_{w_i} = u_{iw_i}\beta_i.$$

Hence β_i is not merely a lower bound: it is attained by an actual MBB good even when $b_i = 0$.

For every $j \in S_i$, impose

$$p_j = u_{ij}\beta_i, \quad y_{ij} \geq \delta,$$

and impose $y_{ij} = 0$ for $j \notin S_i$. Also impose

$$\sum_{j \in P} y_{ij} = b_i.$$

The two possible active-budget regimes are the following closed linear systems:

$$\begin{aligned} \text{budget-limited: } & b_i = m_i, \quad m_i \leq c_i \beta_i, \\ \text{cap-limited: } & b_i = c_i \beta_i, \quad c_i \beta_i \leq m_i. \end{aligned}$$

At equality, either regime may be chosen.

Now suppose buyer i values a good in Z . Impose

$$\beta_i = b_i = 0, \quad y_{ij} = 0 \quad (j \in P),$$

and require a valued witness $f_i \in Z$ with $u_{if_i} > 0$. Free allocation is allowed only on the guessed free support $T_i \subseteq Z$:

$$z_{ij} = 0 \quad (j \notin T_i), \quad z_{ij} \geq \delta \quad (j \in T_i),$$

and

$$\sum_{j \in Z} u_{ij} z_{ij} = c_i.$$

These are the free-good allocation constraints that would be lost if one worked only with spending variables.

For a positive-price good $j \in P$, one of two seller regimes is guessed. In the full-sale regime, impose

$$p_j \leq d_j, \quad \sum_i y_{ij} = p_j.$$

In the capped-earning regime, impose

$$p_j \geq d_j, \quad \sum_i y_{ij} = d_j.$$

At $p_j = d_j$, either regime is valid. For each free good $j \in Z$, impose the physical capacity constraint

$$\sum_i z_{ij} \leq 1.$$

Every constraint above is rational and linear. Maximize δ and accept the guessed regime only if the optimal value satisfies

$$\delta^* > 0.$$

It is not enough merely to write $p_j \geq \delta$, since allowing $\delta = 0$ would fail to enforce the guessed positive-price and positive-support conditions.

If a real equilibrium exists, choose its actual zero set, binding regimes, positive supports, and MBB witnesses. The minimum of 1 and all its finitely many positive prices, spending entries, and free-allocation entries gives a feasible $\delta > 0$. Thus the corresponding linear program has positive optimum.

Conversely, from a feasible solution with $\delta > 0$, define

$$x_{ij} = \begin{cases} y_{ij}/p_j, & j \in P, \\ z_{ij}, & j \in Z. \end{cases}$$

On a full-sale positive-price good,

$$\sum_i x_{ij} = \frac{\sum_i y_{ij}}{p_j} = 1,$$

while on a capped-earning good,

$$\sum_i x_{ij} = \frac{d_j}{p_j} \leq 1.$$

Free-good feasibility holds by the explicit z -constraints.

For every buyer who values no free good, the MBB inequalities and the witnessed equality give

$$\beta_i = \min_{j: u_{ij} > 0} \frac{p_j}{u_{ij}}.$$

Spending occurs only on equality edges and totals $\min\{m_i, c_i \beta_i\}$. Lemma 1 therefore shows that the reconstructed bundle is thrifty and modest. A buyer who values a free good receives raw utility c_i at expenditure zero and is also thrifty and modest.

Finally, rational linear programming has the standard small-solution property: if a rational linear system of polynomial dimension and polynomial input bit length has a solution with $\delta > 0$, then it has such a rational solution of polynomial bit length. Equivalently, one may take a polynomial-bit rational optimal solution of the δ -maximization program. No price normalization is needed or used; earning-capped Fisher markets are not price-homogeneous. The quotients y_{ij}/p_j retain polynomial encoding length.

Thus every yes-instance has a polynomial-bit rational equilibrium, and all equilibrium conditions can be checked by exact rational arithmetic in polynomial time. $\square$

5 Strong NP-hardness

We reduce from EXACT COVER BY 3-SETS. An instance consists of a universe

$$U = \{e_1, \dots, e_{3q}\}$$

and a family $\mathcal{S}$ of three-element subsets of U . The question is whether there is a subfamily $\mathcal{C} \subseteq \mathcal{S}$ such that every element of U belongs to exactly one set in $\mathcal{C}$.

5.1 The construction

All goods have unit physical supply.

Anchor good. Create one good a with earning cap

$$d_a = 24.$$

Create two anchor buyers:

buyer	m_i	c_i	positive utility coefficient
A_1	15	1	$u_{A_1 a} = 2$
A_2	12	1	$u_{A_2 a} = 1.$

Element gadget. For every $e \in U$, create goods H_e and E_e , both with earning cap 12. Create an element buyer Q_e and a blocker D_e^H :

buyer	m_i	c_i	positive utility coefficients
Q_e	12	1	$u_{Q_e H_e} = u_{Q_e E_e} = 2$
D_e^H	0	2	$u_{D_e^H H_e} = 1.$

Set gadget. For every $S \in \mathcal{S}$, create three local goods

$$g_{S,1}, \quad g_{S,2}, \quad g_{S,3},$$

each with earning cap 24.

Create one base buyer B_S , three auxiliary buyers $C_{S,1}, C_{S,2}, C_{S,3}$, and one blocker $D_{S,r}^g$ for each local good:

buyer	m_i	c_i	positive utility coefficients
B_S	36	1	$u_{B_S g_{S,r}} = 2 \quad (r = 1, 2, 3)$
$C_{S,k} \ (k = 1, 2, 3)$	16	2	$u_{C_{S,k} g_{S,r}} = 9 \quad (r = 1, 2, 3)$
$D_{S,r}^g$	0	2	$u_{D_{S,r}^g g_{S,r}} = 1.$

In particular, every auxiliary buyer values all three local goods equally.

Finally, for each occurrence $e \in S$, create a token buyer $T_{S,e}$ with

$$m_{T_{S,e}} = 12, \quad c_{T_{S,e}} = 3,$$

and

$$u_{T_{S,e},a} = u_{T_{S,e},E_e} = u_{T_{S,e},g_{S,1}} = u_{T_{S,e},g_{S,2}} = u_{T_{S,e},g_{S,3}} = 1.$$

All unspecified utility coefficients are zero.

The construction has size polynomial in $|U| + |\mathcal{S}|$.

5.2 The anchor price

Lemma 2. *Every equilibrium of the constructed market satisfies*

$$p_a = 24.$$

At this price, A_1 and A_2 each receive 1/2 unit of a , and no token buyer receives any of a or spends money on it.

Proof. Suppose first that $p_a = 0$. Buyer A_1 can reach its utility cap only by receiving 1/2 unit of a , while A_2 requires one full unit. Since both then value a free good, Lemma 1 requires these cap-reaching allocations. Their combined requirement is

$$\frac{1}{2} + 1 > 1,$$

contradicting the unit supply of a .

Now let $p_a > 0$. Since each anchor buyer values only a , their forced aggregate spending on a is

$$D(p_a) = \min \left\{ 15, \frac{p_a}{2} \right\} + \min \{ 12, p_a \}.$$

The seller of a receives

$$R(p_a) = \min\{p_a, 24\}.$$

Since token spending is nonnegative, a necessary condition is $D(p_a) \leq R(p_a)$.

For $0 < p_a \leq 12$,

$$D(p_a) = \frac{3}{2}p_a > p_a = R(p_a).$$

For $12 < p_a < 24$,

$$D(p_a) = \frac{p_a}{2} + 12 > p_a = R(p_a).$$

At $p_a = 24$,

$$D(24) = 12 + 12 = 24 = R(24).$$

For $24 < p_a < 30$,

$$D(p_a) = \frac{p_a}{2} + 12 > 24 = R(p_a),$$

and for $p_a \geq 30$,

$$D(p_a) = 15 + 12 = 27 > 24 = R(p_a).$$

Thus $p_a = 24$ is the only possibility.

At this price, A_1 spends 12 and receives $1/2$, while A_2 spends its budget 12 and also receives $1/2$. They exhaust both the unit supply and the 24-dollar earning cap. Hence no token can receive or spend on a . $\square$

5.3 Element prices

Lemma 3. *For every $e \in U$, every equilibrium satisfies*

$$p_{H_e} = p_{E_e} = 24.$$

Buyer Q_e spends all 12 dollars on H_e , and the token buyers collectively spend exactly 12 dollars on E_e .

Proof. The blocker D_e^H rules out $p_{H_e} = 0$. Indeed, if H_e were free, the blocker would have to receive two units to attain its utility cap, contradicting unit supply. Thus $p_{H_e} > 0$.

Only Q_e can make a positive payment for H_e . Since the seller of H_e must receive positive revenue, Q_e must buy H_e , and therefore H_e is MBB for Q_e . Because Q_e values H_e and E_e equally,

$$p_{H_e} \leq p_{E_e}.$$

When H_e is MBB, the active expenditure of Q_e is

$$\min\left\{12, \frac{p_{H_e}}{2}\right\}.$$

Consequently,

$$\min\{p_{H_e}, 12\} \leq \min\left\{12, \frac{p_{H_e}}{2}\right\}.$$

If $0 < p_{H_e} < 12$, this would require $p_{H_e} \leq p_{H_e}/2$. If $12 \leq p_{H_e} < 24$, it would require $12 \leq p_{H_e}/2 < 12$. Both are impossible, so

$$p_{H_e} \geq 24.$$

It follows that $p_{E_e} \geq 24$.

At $p_{H_e} \geq 24$, good H_e must earn 12, and Q_e 's entire active budget is 12. Thus all of Q_e 's spending is needed for H_e , leaving no money from Q_e for E_e .

Since $p_{E_e} \geq 24$, good E_e must also earn 12. Therefore some occurrence token must buy E_e . Every such token values both E_e and the anchor good a at coefficient 1. Since $p_a = 24$, MBB optimality for a token buying E_e implies

$$\frac{1}{p_{E_e}} \geq \frac{1}{24}, \quad \text{hence} \quad p_{E_e} \leq 24.$$

Combining the inequalities gives

$$24 \leq p_{H_e} \leq p_{E_e} \leq 24.$$

Thus both prices equal 24, and E_e receives exactly 12 dollars of aggregate token spending. $\square$

5.4 The two states of a set gadget

Lemma 4. *In every equilibrium, the three local goods of a set S have a common price t_S , and*

$$t_S \in \left\{ \frac{216}{11}, \frac{432}{7} \right\}.$$

Proof. Every local price is positive. If $p_{g_{S,r}} = 0$, its blocker $D_{S,r}^g$ would require two units of that good to attain its cap, which is impossible.

All positive-budget buyers who value the local goods value the three goods symmetrically. If the positive local prices were unequal, any strictly more expensive local good would be MBB for none of these buyers. Its blocker has budget zero, so the good would receive no revenue, contradicting its positive price. Hence all three local prices are equal; denote their common value by t .

The base buyer's active spending is

$$s_B(t) = \min \left\{ 36, \frac{t}{2} \right\},$$

and every auxiliary buyer's active spending is

$$s_C(t) = \min \left\{ 16, \frac{2t}{9} \right\}.$$

A token compares its local bang per buck $1/t$ with its external bang per buck

$$\frac{1}{p_a} = \frac{1}{p_{E_e}} = \frac{1}{24}.$$

Suppose first that $0 < t < 24$. Every token strictly prefers the local goods, and its active expenditure is

$$\min\{12, 3t\}.$$

If $t < 4$, each token must attain its cap by receiving three total units of local goods. The three tokens would then require nine units, whereas the three local goods have only three units in total. Hence $t \geq 4$, after which every token spends exactly 12 dollars locally.

For $4 \leq t < 24$, the local goods sell their full units, so their total revenue is $3t$. Base and auxiliary spending is

$$\frac{t}{2} + 3 \left(\frac{2t}{9} \right) = \frac{7t}{6},$$

and the three tokens contribute 36. Market clearing gives

$$3t = \frac{7t}{6} + 36,$$

whose unique solution is

$$t = \frac{216}{11}.$$

This value lies in $[4, 24)$.

At $t = 24$, base and auxiliary buyers contribute

$$\frac{24}{2} + 3 \left(\frac{2 \cdot 24}{9} \right) = 12 + 16 = 28.$$

The three token budgets contribute at most 36, so local revenue is at most 64, while the three local sellers require 72. Thus $t = 24$ is impossible.

Now suppose that $24 < t < 72$. Tokens strictly prefer their external goods, so they spend nothing locally. Each local good reaches its 24-dollar earning cap, for total required revenue 72. The base and auxiliary buyers are still below their budget bounds, so

$$72 = \frac{7t}{6}.$$

Thus

$$t = \frac{432}{7},$$

which lies strictly between 24 and 72.

Finally, if $t \geq 72$, the base buyer spends 36, and each of the three auxiliary buyers spends 16. Their local spending alone is

$$36 + 3(16) = 84,$$

which exceeds the total local earning capacity 72. This is impossible. The two displayed prices are therefore the only states. $\square$

Call a set *low* if its common local price is $216/11$, and *high* if its common local price is $432/7$.

5.5 An equilibrium yields an exact cover

Lemma 5. *If the constructed market has a thrifty-and-modest equilibrium, then the high sets form an exact cover of U .*

Proof. Fix a high set S and an element $e \in S$. Token $T_{S,e}$ faces local price

$$\frac{432}{7} > 24.$$

It therefore strictly prefers its external goods a and E_e to the local goods. Its utility cap is 3, while reaching that cap at external price 24 would cost 72. Hence the token is budget-limited and must spend all 12 dollars.

By Lemma 2, the anchor good is already fully allocated to A_1 and A_2 , and its earning cap is exhausted. Therefore $T_{S,e}$ must spend all 12 dollars on E_e .

A token belonging to a low set spends all its money locally because $216/11 < 24$, so it spends nothing on E_e . By Lemma 3, the aggregate token flow into E_e is exactly 12. Consequently,

$$12 \cdot |\{S \in \mathcal{S} : S \text{ is high and } e \in S\}| = 12.$$

Exactly one high set contains e . Since this holds for every $e \in U$, the high sets form an exact cover. $\square$

5.6 An exact cover yields an equilibrium

Lemma 6. *If the EXACT COVER BY 3-SETS instance has an exact cover, then the constructed market has a thrifty-and-modest equilibrium.*

Proof. Let $\mathcal{C} \subseteq \mathcal{S}$ be an exact cover. Set

$$p_a = 24, \quad p_{H_e} = p_{E_e} = 24 \quad (e \in U).$$

For the local goods of S , set

$$t_S = \begin{cases} 432/7, & S \in \mathcal{C}, \\ 216/11, & S \notin \mathcal{C}. \end{cases}$$

Allocate the anchor good by

$$x_{A_1 a} = x_{A_2 a} = \frac{1}{2}.$$

For every $e \in U$, allocate

$$x_{Q_e H_e} = \frac{1}{2}.$$

Because $\mathcal{C}$ is an exact cover, there is a unique selected set $S(e) \in \mathcal{C}$ containing e . Allocate

$$x_{T_{S(e),e}, E_e} = \frac{1}{2}.$$

For every set S and every local good $g_{S,r}$, allocate

$$x_{B_S g_{S,r}} = \frac{1}{6}$$

and

$$x_{C_{S,k} g_{S,r}} = \frac{2}{27} \quad (k = 1, 2, 3).$$

If $S \notin \mathcal{C}$, additionally allocate

$$x_{T_{S,e}, g_{S,r}} = \frac{11}{54} \quad (e \in S, r = 1, 2, 3).$$

Tokens of selected sets receive no local goods. All blockers and all unlisted buyer–good pairs receive zero.

For a low local good, the total allocation is

$$\frac{1}{6} + 3 \left(\frac{2}{27} \right) + 3 \left(\frac{11}{54} \right) = 1.$$

Its price 216/11 is below its earning cap, so selling the full unit is correct.

For a high local good, the total allocation is

$$\frac{1}{6} + 3 \left(\frac{2}{27} \right) = \frac{7}{18} = \frac{24}{432/7}.$$

It therefore earns exactly 24, as required by its earning cap.

The anchor good sells one unit and earns 24. Every H_e and E_e sells 1/2 unit at price 24, earning its cap 12.

It remains to verify buyer optimality. The relevant raw utilities and expenditures are:

buyer type	raw utility	expenditure
A_1	1	12
A_2	1/2	12
Q_e	1	12
B_S	1	$t_S/2$
$C_{S,k}$	2	$2t_S/9$
low token	11/18	12
high token	1/2	12.

Buyer A_1 reaches its cap at minimum cost; A_2 exhausts its budget. Buyer Q_e receives an MBB good and reaches its cap. Every base buyer reaches utility 1, and every auxiliary buyer reaches utility 2, using only equally priced local MBB goods. Their expenditures lie within their budgets at both local prices.

For a low token,

$$\frac{1}{216/11} > \frac{1}{24},$$

so its local goods are strictly MBB. It receives total local quantity

$$3 \left(\frac{11}{54} \right) = \frac{11}{18}$$

and spends

$$\frac{216}{11} \cdot \frac{11}{18} = 12.$$

For a high token,

$$\frac{1}{24} > \frac{1}{432/7},$$

so E_e is MBB, and the token spends its budget 12 on 1/2 unit of E_e . All token utilities remain below the cap 3, so the tokens are modest and budget-limited.

Every blocker has budget zero and its sole valued good has positive price, so its zero allocation is optimal. Thus all buyers are thrifty and modest, all physical supply constraints hold, and every seller receives the required revenue. $\square$

6 Main theorem

Theorem 1. *The problem THRIFTY-AND-MODEST EQUILIBRIUM EXISTENCE is strongly NP-complete.*

The result holds even when every good has unit supply and the numerical parameters are restricted to

$$u_{ij} \in \{0, 1, 2, 9\}, \quad m_i \in \{0, 12, 15, 16, 36\}, \quad c_i \in \{1, 2, 3\}, \quad d_j \in \{12, 24\}.$$

Proof. Proposition 1 establishes membership in NP.

The construction from EXACT COVER BY 3-SETS is polynomial. By Lemma 5, an equilibrium implies that the high sets form an exact cover. By Lemma 6, every exact cover yields an equilibrium. Hence the reduction is correct.

The source problem is strongly NP-complete, and all numerical values used in the constructed market belong to fixed finite sets independent of the input size. The reduction therefore establishes strong NP-hardness. Together with membership in NP, this proves strong NP-completeness. $\square$

7 Scope of the result

The theorem concerns the general model with nonnegative budgets. The zero-budget blockers are legal because the model permits $m_i = 0$; they are used to rule out zero prices for H_e and for the local goods. Consequently, the reduction does not establish hardness for the variant in which every buyer must have a strictly positive budget.

The construction also lies outside the money-clearing subclass. The two anchor buyers value only the anchor good and satisfy

$$m_{A_1} + m_{A_2} = 15 + 12 = 27 > 24 = d_a.$$

Thus the relevant neighborhood money-clearing inequality already fails inside the anchor gadget. The theorem therefore does not contradict the guaranteed existence of equilibrium in money-clearing markets.

Finally, the theorem is an equilibrium-existence result. It does not establish hardness of finding an exact equilibrium under a promise that one exists, nor does it establish search hardness for money-clearing instances.

References

- [1] Jugal Garg, Martin Hoefer, Kurt Mehlhorn. *Satiation in Fisher Markets and Approximation of Nash Social Welfare*. Mathematics of Operations Research, 2023. <https://doi.org/10.1287/moor.2019.0129>